

$$\equiv \forall x \exists y (\neg P(x) \vee Q(y)) \quad (\text{Ex-49(b)})$$

52) Express the quantification $\exists! x P(x)$, introduced in Section 1.4, using universal quantifications, existential quantifications and logical operators.

Soln: **Claim:** $\exists! x P(x)$ and $\exists x (P(x) \wedge \forall y ((x \neq y) \rightarrow \neg P(y)))$ are logically equivalent.

Proof: Let $\exists! x P(x)$ be True. By defn. of uniqueness quantifier, there exists exactly one element a in the domain where $P(a)$ is True. For any other element b in the domain, $P(b)$ is False. By defn. of negation, $\neg P(b)$ is True. $(a \neq b)$ is True. By defn. of conditional, $\boxed{(a \neq b) \rightarrow \neg P(b)}$ is True.

This holds true for any arbitrary b , since when $a=b$, $a \neq b$ is False and the above conditional is True by defn. By defn. of universal quantifier, $\forall y ((a \neq y) \rightarrow \neg P(y))$ is True. By defn. of AND, $P(a) \wedge \forall y ((a \neq y) \rightarrow \neg P(y))$ is True. By defn. of existential quantifier, $\exists x (P(x) \wedge \forall y ((x \neq y) \rightarrow \neg P(y)))$ is True.

Let, $\exists x (P(x) \wedge \forall y ((x \neq y) \rightarrow \neg P(y)))$ is True. By defn. of existential quantifier, there exists some element a in the domain s.t. $(P(a) \wedge \forall y ((a \neq y) \rightarrow \neg P(y)))$ is True.

By defn. of AND, $P(a)$ is True and $\forall y ((a \neq y) \rightarrow \neg P(y))$ is True.

By defn. of universal quantifier, for all elements b in the domain, $(a \neq b) \rightarrow \neg P(b)$ is True. If $a=b$ is True, then $a \neq b$ is False and the conditional is True by defn. If $(a \neq b)$ is True, by defn. of conditional, $\neg P(b)$ is True. $\therefore$ For all elements which are not a , $\neg P(b)$ is True.

By defn. of negation, $\neg(\neg P(b))$ is False. By double-negation law, $P(b)$ is False. By defn. of uniqueness quantifier, $\exists! x P(x)$ is True.

$$\boxed{\exists! x P(x) \equiv \exists x (P(x) \wedge \forall y ((x \neq y) \rightarrow \neg P(y)))} \quad \square$$